Lecture 9

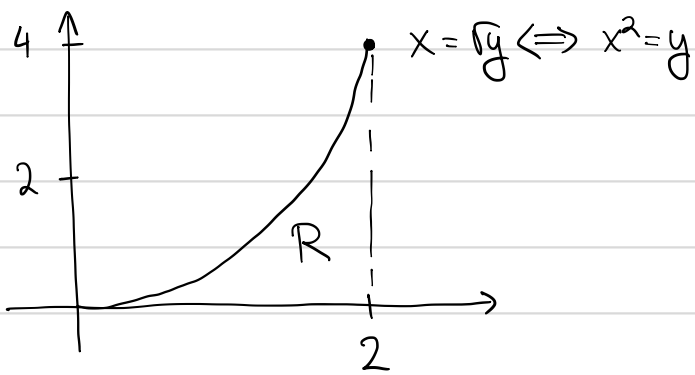
Review for Midterm 1.

Good exams from archive:
Arms, Folland.

Arms (1a)

$$\int_0^4 \int_{\sqrt{y}}^2 f(x,y)\,dx\,dy =$$

$$= \int_0^2 \int_0^{x^2} f(x,y)\,dy\,dx$$



1b) $\bar{x} = \dfrac{1}{\text{mass}} \iint_R x\rho\,dA \qquad \rho = ke^y$

$$= \frac{1}{m}\int_0^2\int_0^{x^2} x\cdot k e^y\,dy\,dx$$

$$= \frac{k}{m}\int_0^2 x\, e^y\Big|_{y=0}^{x^2}\,dx$$

$$= \frac{k}{m}\int_0^2 x\left(e^{x^2}-1\right)dx$$

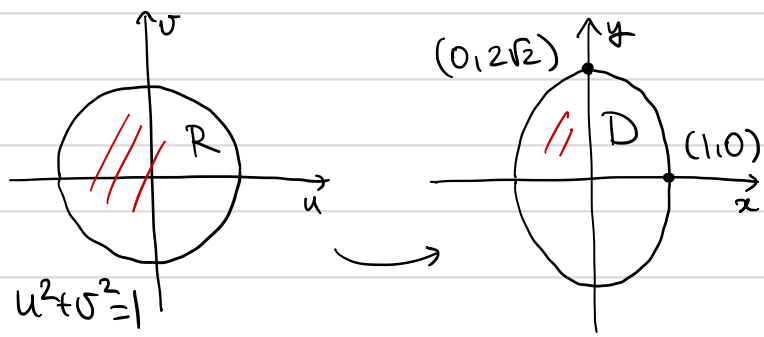
$$= \frac{k}{m}\left[\frac{1}{2}e^{x^2} - \frac{x^2}{2}\right]_0^2$$

$$\bar{x} = \frac{k}{2m}\left(e^4 - 1 - 4 + 0\right)$$

$$= \frac{k}{2m}\left(e^4 - 5\right) \;//$$

Example: $\iint_D e^{8x^2+y^2}\,dA = I$

$D =$ interior of $x^2 + \left(\frac{y}{2\sqrt{2}}\right)^2 = 1.$



$x = u, \quad y = 2\sqrt{2}\,v$

$$I = \iint_R e^{8u^2+8v^2}\left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\,dv$$

$$\frac{\partial(x,y)}{\partial(u,v)} = \det\begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v}\end{bmatrix}$$

$$= \det\begin{bmatrix} 1 & 0 \\ 0 & 2\sqrt{2}\end{bmatrix} = 2\sqrt{2}$$

$$I = \iint_R e^{8u^2+8v^2}\,2\sqrt{2}\,du\,dv$$

$$= \int_{\theta=0}^{2\pi}\int_{r=0}^{1} e^{8r^2}\,2\sqrt{2}\,r\,dr\,d\theta$$

$u = r\cos\theta$
$v = r\sin\theta$

$$I = 2\sqrt{2}\cdot 2\pi \int_{r=0}^{1} e^{8r^2}\, r\,dr$$

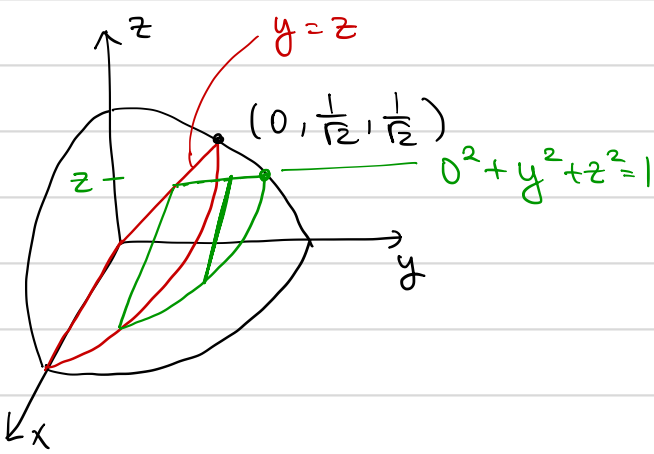
$$= 2\sqrt{2}\cdot 2\pi \left[\frac{e^{8r^2}}{16}\right]_0^1$$

$$= \frac{\pi\sqrt{2}}{4}(e-1) \;//$$

Arms (3a)

E inside $x, y, z \geq 0$,
" $x^2+y^2+z^2 \leq 1$,
below $z=y$.

$$\int_0^{1/\sqrt{2}} \int_z^{\sqrt{1-z^2}} \int_0^{\sqrt{1-y^2-z^2}} f(x,y,z)\,dx\,dy\,dz$$



3b) Why $\underline{dz}\,dx\,dy$ and $\underline{dz}\,dy\,dx$ bad ideas?

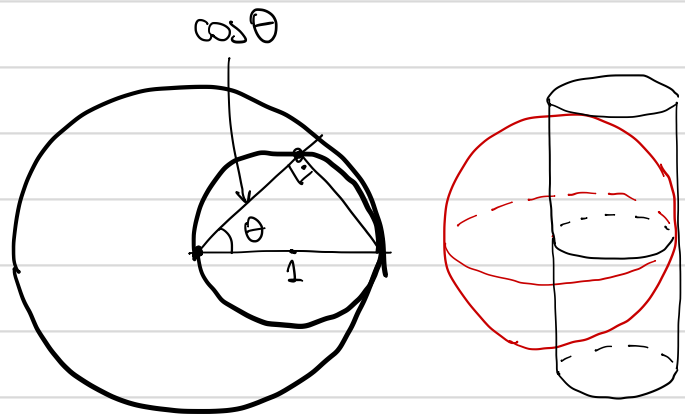
Wong, Sample 2 (3)

Vol. between:

$x^2+y^2+z^2=1$ (sphere)

$r=\cos\theta$ (cylinder)

$$\left.\begin{aligned} r^2 &= r\cos\theta \\ x^2+y^2 &= x \\ (x-\tfrac{1}{2})^2+y^2 &= \tfrac{1}{4} \end{aligned}\right\} \quad \begin{aligned}&\text{Center: } (\tfrac{1}{2}, 0) \\ &\text{Radius: } \tfrac{1}{2}\end{aligned}$$

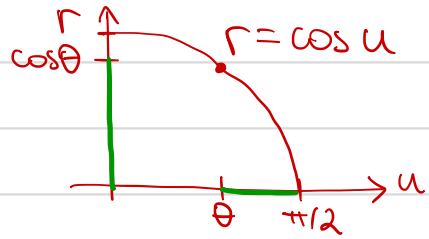


$$\int_{\theta=-\frac{\pi}{2}}^{\pi/2} \int_{r=0}^{\cos\theta} \int_{z=-\sqrt{1-r^2}}^{\sqrt{1-r^2}} dz\,dr\,d\theta$$

$$= \int_{-\pi/2}^{\pi/2} \int_0^{\cos\theta} 2\sqrt{1-r^2}\,dr\,d\theta$$

$r = \cos u \longrightarrow dr = -\sin u\,du$

$$= \int_{-\pi/2}^{\pi/2} \int_{\theta}^{\pi/2} 2\underbrace{\sqrt{\sin^2 u}}_{\sin(u)\ \text{since}\ 0\leq u\leq\pi}(-\sin u)\,du\,d\theta$$



$$\cdots = \frac{\pi}{2}(\pi-1)$$